

Text Summarization (Extractive)-

15/10/25(Wednesday):

This project successfully developed an interactive, command-line chatbot designed around the principles of rule-based Natural Language Processing (NLP). The primary function of the bot is to perform a simplified extractive text summarization. The system determines a user's intent using keyword matching and, upon receiving a text for summarization, returns the text's first sentence as the summary. The project fulfills all requirements, demonstrating mastery of fundamental programming concepts like data structures (dictionaries), text preprocessing, conditional logic, and conversational loops.

2. Project Goals

The main objectives of this assignment were to:

1. Define Intents and Queries: Establish a mapping between user goals (Intents) and the keywords that trigger them.
2. Implement Text Preprocessing: Clean and normalize user input for reliable rule matching.
3. Develop Rule-Based Logic: Use explicit rules (keywords/regex) to match input to defined intents.
4. Create an Extractive Summarization Function: Design a simple method to extract the most important part of a text.
5. Build an Interactive Loop: Construct a continuous environment for user-chatbot interaction.

3. Technical Implementation

A. Intent and Rule Mapping

The core intelligence is stored in a dictionary, `INTENT_DATA`, which directly links intents (user goals) to their corresponding triggering keywords and response templates. This demonstrates a rule-based approach where responses are hard-coded, not learned.

Intent	Purpose	Triggering Keywords
Summarize	To trigger the main summarization action.	summarize, text, extract, shorten
Greeting	To initiate the conversation.	hello, hi, hey
Function_Query	To ask the bot's purpose.	what, do, purpose, function

B. Preprocessing

The `preprocess_text` function ensures all input is standardized before matching, drastically reducing the chance of matching failure due to variations in case or punctuation.

Steps:

1. Lowercasing: Converts all characters to lowercase ("HELLO" → "hello").
2. Punctuation Removal: Removes common punctuation using the `re` module ("hello!" → "hello").

C. Extractive Summarization Logic

This project utilizes the most basic method of extractive summarization: Position-Based Extraction.

The `generate_simple_summary` function assumes that the first sentence of any document is the most relevant or representative. The logic uses regular expressions to find the first sentence-ending punctuation (., ?, or !) and returns all text up to that point.

D. The Conversational Loop

The `start_chatbot` function contains the continuous `while True` loop. This loop performs the following actions for every user input:

1. Exit Check: Breaks the loop if input matches an exit command (bye, stop).

2. Intent Detection: Calls `get_response` to match the processed input to an intent.

3. Action Flow:

- If the intent is Summarize, the loop pauses, prompts the user to paste the text, executes the `generate_simple_summary` function, and prints the result.
- For all other intents, it simply prints the pre-defined response.

4. Mathematical Concept (Position Scoring)

While no complex algorithms were used, the principle governing the extraction is the simplest form of Sentence Position Scoring. This concept assigns relevance based purely on a sentence's location in the document.

The selection rule used in this project can be mathematically represented as:

Select Sentence S_i if $i=1$

Where:

- S_i is the sentence at index i in the document.
- $i=1$ is the first sentence.

This scoring mechanism ensures the system reliably extracts the initial statement, fulfilling the requirement for basic extractive summarization.

5. Conclusion and Future Work

The Rule-Based Extractive Text Summarization Chatbot is a successful introductory NLP project. It effectively maps user intent to specific actions and delivers a functional, though basic, summarization utility.

Future Enhancements:

- Advanced Summarization: Implement a weighted scoring system, such as using keyword frequency (e.g., TF-IDF), to identify truly relevant sentences beyond just the first one.
- Context Management: Add basic memory to allow the bot to refer to previous topics or the user's name.
- Response Variety: Introduce the random library to select from multiple response templates for each intent, making the conversation feel more natural.

A short summary:-

This project created a Rule-Based Extractive Text Summarization Chatbot, which serves as an introductory demonstration of Natural Language Processing (NLP).

The chatbot works by:

1. Defining Intents: Using pre-set rules (keywords) to classify the user's goal (e.g., Greeting, Summarize).
2. Preprocessing: Cleaning user input (lowercasing, removing punctuation) to enable reliable keyword matching.
3. Extractive Summarization: Performing the core function by implementing a simple Position Scoring method, which dictates that the first sentence of any provided text is the most important and is therefore extracted as the summary.
4. Interaction: Running within a continuous conversation loop that handles various user inputs and executes the summarization action upon request.

The project successfully built an intent-driven, rule-based system, laying a foundation for more advanced NLP techniques.